

REPRODUCIBILITY_REPORT.md

DSN Bootcamp Qualification Hackathon 2026 ML Track

Reproducibility Report

Report Date: 2026-09-22

Summary

Both final submission options are **BYTE-IDENTICALLY REPRODUCIBLE** under the exact conditions documented in `REPRODUCIBILITY_CERTIFICATE.md`.

Canonical Reproduction

- **Status:** BYTE-IDENTICAL
- **Script:** `_exec_notebook.py` (archived)
- **Verification:** `CODE/final/run_notebook_repro.py` confirms SHA256 match
- **Evidence:** `FINAL_RELEASE/reproduction/TRUE_NOTEBOOK_OUTPUT.csv` matches canonical hash

E3 Reproduction

- **Status:** BYTE-IDENTICAL
- **Script:** `ARCHIVE/REPORTS/PHASE11C_production_train.py`
- **Verification:** 2-run reproducibility check confirmed identical hashes
- **Critical requirement:** CatBoost 1.2.10 exactly

Environment

```
numpy==1.26.4  
pandas==2.1.4  
scikit-learn==1.4.2  
catboost==1.2.10  
scipy==1.12.0
```

Reproducibility Protocol

1. Install pinned package versions
2. Ensure data files at documented paths
3. Execute authoritative script
4. Compute SHA256 of output
5. Compare against recorded hash

Limitations

- CatBoost version sensitivity (E3)
- Working directory requirements (E3)
- No private leaderboard validation
- No guarantee of identical results on different hardware/OS

Report generated as part of FINAL COMPETITION RELEASE / REPRODUCIBILITY WAR ROOM.